

Part 0: Introduction

The Singapore public housing (HDB) resale market is shaped by a complex interplay of physical characteristics, location, and lease tenure. The primary objective of this report is to identify the best-fitting linear regression model for predicting the *resale price* of HDB flats based on the transaction data for July 2020. This goal is practically significant: an accurate pricing model supports buyers, sellers, valuers, and policymakers in making evidence-based decisions in one of the world's most densely transacted public housing markets. The dataset contains 2,456 observations extracted from data.gov.sg, each representing one completed resale transaction in July 2020. The response variable is *resale_price* (in SGD). Ten feature variables are available: *month*, *town*, *flat_type*, *block*, *street_name*, *storey_range*, *floor_area_sqm*, *flat_model*, *lease_commence_date*, and *remaining_lease*. The report proceeds in two main parts. Part I presents an Exploratory Data Analysis (EDA) of the variables, examining distributions, relationships, and data-quality issues. Part II covers model building, beginning with an initial OLS model (M₀) using all available regressors, followed by diagnostic checks and remediation, and concluding with a final selected model (M_N). Model adequacy is evaluated through residual diagnostics, Variance Inflation Factor (VIF) analysis, and information criteria (AIC/BIC).

Part I: Exploratory Data Analysis

Before analysis, the following pre-processing steps were applied. (i) Two observations were removed: the sole *MULTI-GENERATION* flat and the sole *Type S2* flat model, each being a singleton class unusable for regression. This left **2,454 observations**. (ii) The *month* column was dropped, as every record has the value 2020-07. (iii) Three high-cardinality or identifier columns (*block*, *street_name*, and *storey_range*) were removed or transformed. (iv) *storey_range* (a string such as '04 TO 06') was converted to a numeric *average_storey* by averaging the lower and upper bound of each range, providing an ordinal continuous proxy. (v) *remaining_lease* (stored as a string like '56 years 7 months') was parsed into a numeric *remaining_lease_years* variable in terms of years. (vi) *town* (26 categories) was recoded into *region* (5 categories: Central, North, North-East, East, West) using the Urban Redevelopment Authority's official planning regions, substantially reducing dimensionality while preserving geographic meaning. Thereafter, *town*, *block*, *street_name*, *storey_range*, *remaining_lease* were dropped.

The raw *resale_price* distribution is right-skewed (Figure 1), with a long tail of high-value flats (mainly Executive and premium Central-region units). A log transformation, $\ln(\text{resale_price})$, produced a visibly more symmetric, approximately normal distribution, justifying its use as the response in all subsequent models.

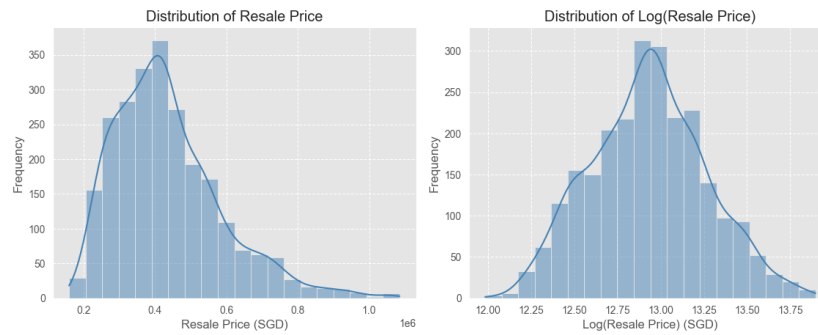


Figure 1: Histogram of resale_price and log(resale_price)

Several important patterns emerged from descriptive plots (Figure 2):

- **flat_type:** Box plots reveal a clear positive gradient, 1 ROOM < 2 ROOM < 3 ROOM < 4 ROOM < 5 ROOM < EXECUTIVE, reflecting the strong size-price relationship.
- **floor_area_sqm:** A positive linear association with *resale_price* is evident, though residual variance appears to fan out at larger floor areas, suggesting mild heteroscedasticity.
- **average_storey:** A weaker but positive relationship exists. Most observations cluster below storey 20, with sparse data at higher floors.
- **lease_commence_date / remaining_lease_years:** Newer leases command higher prices. A notable price spike occurs at 2018 commencement. Critically, these two variables are near-perfectly collinear (Chi-squared test of independence, $p < 0.05$), as *remaining_lease* is algebraically determined by *lease_commence_date* given a fixed 99-year maximum lease.

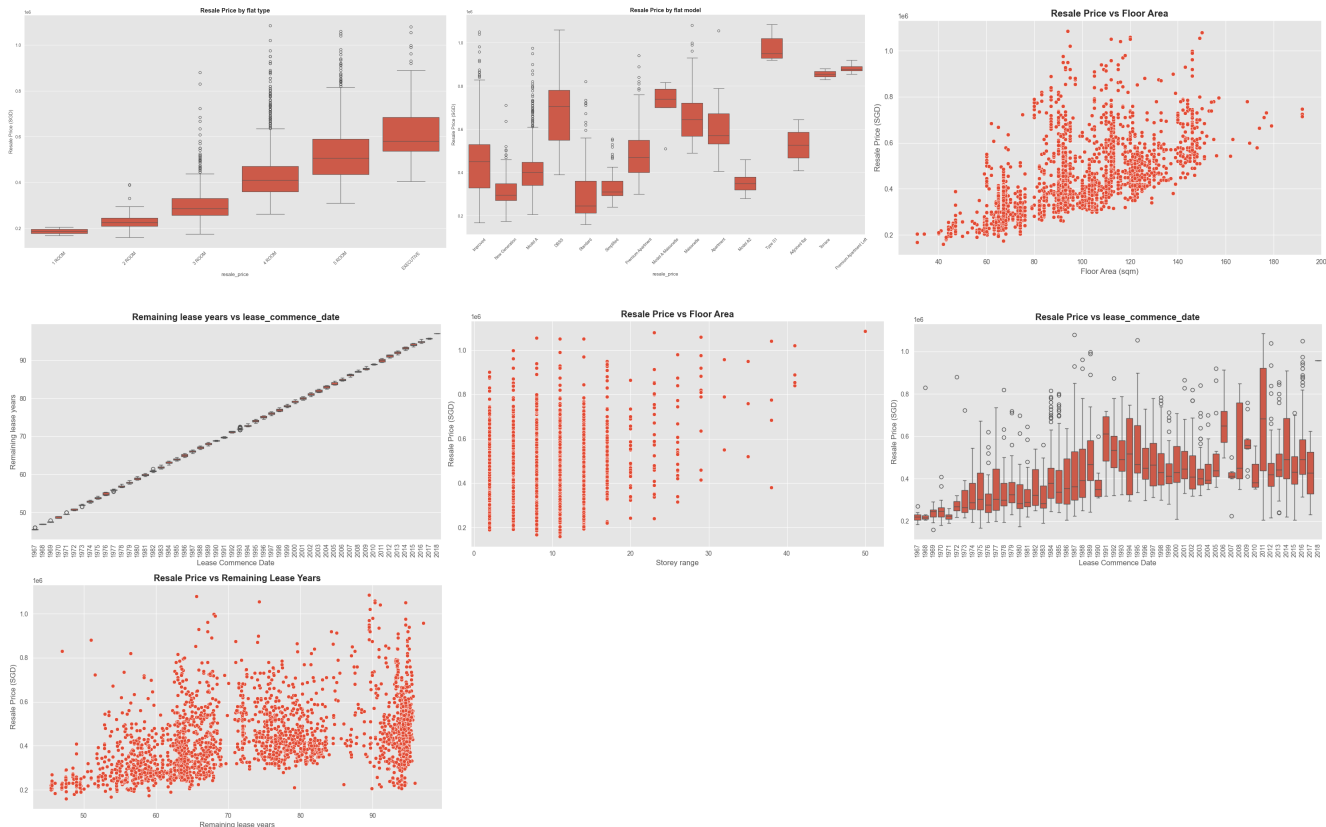


Figure 2: Plots between predictor and response variables

Part II: Model Building

2.1 Initial Model M₀: Full OLS with Log-Transformed Response

The initial model M₀ regresses $\ln(\text{resale_price})$ on all available post-processing predictors: *flat_type*, *flat_model*, *region*, *floor_area_sqm*, *average_storey*, *remaining_lease_years*, and *lease_commence_date*. Categorical predictors are one-hot encoded yielding 27 parameters (excluding intercept).

OLS Regression Results						
Dep. Variable:	resale_price	R-squared:	0.860			
Model:	OLS	Adj. R-squared:	0.859			
Method:	Least Squares	F-statistic:	552.5			
Date:	Thu, 09 Apr 2026	Prob (F-statistic):	0.00			
Time:	21:05:17	Log-Likelihood:	1615.9			
No. Observations:	2454	AIC:	-3176.			
Df Residuals:	2426	BIC:	-3013.			
Df Model:	27					
Covariance Type:	nonrobust					

	coef	std err	t	Pr(> t)	[0.025	0.975]
const	14.2879	17.051	0.840	0.424	-20.717	49.292
flat_type_2 ROOM	0.0011	0.001	0.909	0.358	-0.000	0.219
flat_type_3 ROOM	0.7391	0.001	2.622	0.009	0.000	0.418
flat_type_4 ROOM	0.3372	0.005	3.355	0.001	0.132	0.543
flat_type_5 ROOM	0.3684	0.100	3.680	0.000	0.165	0.550
flat_type_EXECUTIVE	0.2529	0.106	2.763	0.006	0.045	0.501
flat_model_Apartment	0.2180	0.005	3.200	0.000	0.007	0.209
flat_model_S005	0.6663	0.001	4.879	0.000	0.267	0.626
flat_model_Supervised	0.0880	0.000	0.944	0.345	-0.001	0.201
flat_model_Maisonette	0.2051	0.005	0.854	0.402	0.182	0.175
flat_model_Model A	0.0012	0.000	1.000	0.277	-0.079	0.275
flat_model_Model A-Maisonette	0.3541	0.106	3.350	0.001	0.147	0.561
flat_model_Model A2	0.0715	0.005	0.753	0.450	-0.114	0.257
flat_model_Row Generation	0.3080	0.000	2.209	0.028	0.021	0.236
flat_model_Premium Apartment	0.1145	0.000	1.107	0.265	0.003	0.220
flat_model_Premium Apartment Left	0.3370	0.110	2.880	0.004	0.101	0.534
flat_model_Singleflat	0.1620	0.002	7.700	0.000	0.418	0.340
flat_model_Standard	-0.0053	0.001	-0.850	0.404	-0.104	0.174
flat_model_Terrace	0.0710	0.127	0.560	0.580	0.014	0.121
flat_model_Type S1	0.6111	0.107	0.841	0.400	0.222	0.640
region_East	-0.2108	0.010	-21.645	0.000	-0.236	-0.187
region_North	-0.4193	0.010	-40.641	0.000	-0.447	-0.400
region_North-East	-0.2910	0.010	-29.741	0.000	-0.311	-0.272
region_South	-0.1503	0.010	-15.700	0.000	-0.169	-0.131
floor_area_sqm	0.0001	0.000	16.001	0.000	0.000	0.000
lease_commence_date	-0.0015	0.000	-16.172	0.000	-0.002	-0.001
remaining_lease_years	0.0109	0.000	1.173	0.241	-0.007	0.029
average_storey	0.0022	0.000	10.064	0.000	0.000	0.010

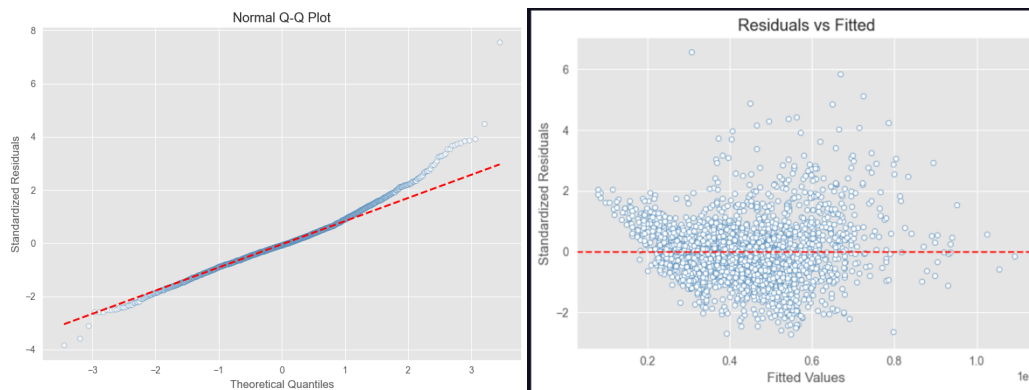
Table 2: M₀ overall model statistics.

	df	sum_sq	mean_sq	F	PR(>F)
flat_type	5.0	148.508804	29.701761	1871.659958	0.000000e+00
flat_model	14.0	38.144487	2.724606	171.691381	0.000000e+00
region	4.0	26.857717	6.714429	423.110546	7.942391e-277
floor_area_sqm	1.0	0.000025	0.000025	0.001594	9.681515e-01
lease_commence_date	1.0	17.541336	17.541336	1105.369330	4.772901e-200
remaining_lease_years	1.0	0.021004	0.021004	1.323550	2.500693e-01
average_storey	1.0	5.646938	5.646938	355.842480	3.564084e-74
Residual	2426.0	38.498698	0.015869	NaN	NaN

Table 3: M₀ ANOVA

The F-statistic of 552.5 ($p < 0.001$) indicates that the model as a whole is highly significant. The R² of 0.860 indicate that the model explains approximately **86%** of the variation in log resale price.

Standardised residuals were examined against each numerical predictor and via a Normal Q-Q plot:



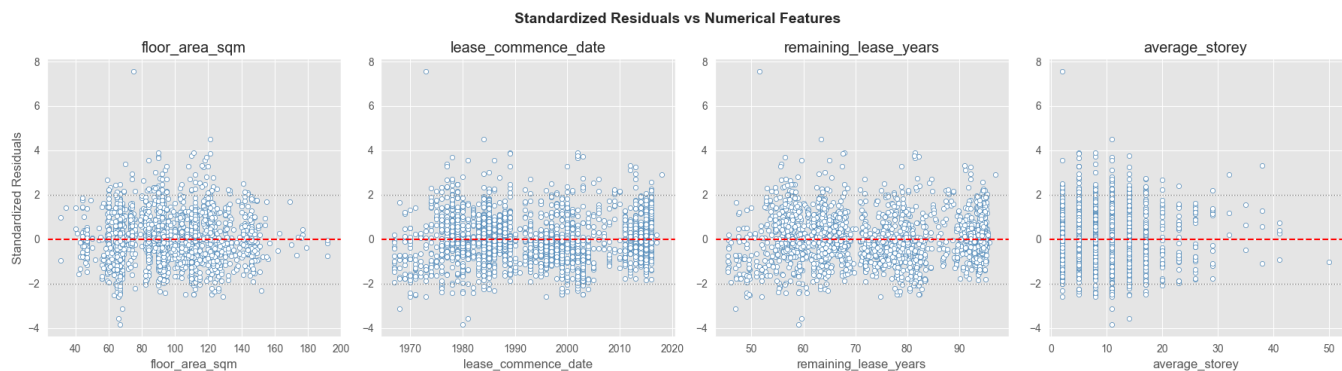


Figure 3: QQPlot and Residual Plots

- **Normality:** The Q-Q plot shows a slightly light-tailed distribution with a few outliers.
- **Linearity:** Residual-vs-predictor plots show no systematic curvature, supporting the linearity assumption.
- **Homoscedasticity:** A slight funnel pattern is visible in the residuals vs. *floor_area_sqm* plot, indicating mild heteroscedasticity for larger flats.

Several observations had a standardised residual exceeding 3, but were still kept as they might be useful. Cook's Distance was computed for all observations: no point exceeded the threshold of 1, so no influential observations were removed.

VIF values for M_0 revealed severe multicollinearity (Table 4). *lease_commence_date* had $VIF = 2,841$, and several *flat_type* and *flat_model* dummies had VIFs in the hundreds. This aligns with the EDA finding that *lease_commence_date* and *remaining_lease_years* are near-perfectly collinear ($r \approx 1$), and that *flat_type* and *flat_model* are structurally associated.

	Feature	VIF
0	lease_commence_date	2841.313322
1	flat_type_4 ROOM	554.947690
2	flat_model_Model A	423.216596
3	flat_type_5 ROOM	375.738707
4	floor_area_sqm	353.081603
5	flat_type_3 ROOM	321.749070
6	flat_model_Improved	314.860396
7	flat_model_New Generation	158.175918
8	flat_type_EXECUTIVE	136.410852
9	flat_model_Premium Apartment	132.353089
10	remaining_lease_years	102.220643
11	flat_model_Apartment	56.173973
12	flat_model_Simplified	45.880733
13	flat_model_Maisonette	41.014230
14	flat_model_DBSS	40.532823
15	flat_model_Standard	37.337610
16	flat_type_2 ROOM	32.522978
17	flat_model_Model A2	11.288073
18	region_North-East	3.996984
19	average_storey	3.905308
20	flat_model_Type S1	3.582640
21	flat_model_Model A-Maisonette	3.504604
22	region_West	3.415953
23	flat_model_Premium Apartment Loft	3.066561
24	region_North	2.755122
25	region_East	2.620785
26	flat_model_Terrace	2.024186

Table 4: VIF comparison between M_0 and M_n (selected features).

2.2 Model M_2 : Ridge Regression

To address the multicollinearity in M_0 , Ridge Regression was explored as an intermediate step. Using leave-one-out cross-validation (RidgeCV) across $\alpha \in [0.1, 100]$, the optimal regularisation parameter was found to be $\alpha = 0.4$. A Ridge trace plot confirmed that coefficient shrinkage was stable and monotone. However, the Ridge test R^2 (0.8573) and RMSE (0.1252) were virtually identical to M_0 test R^2 (0.8572) and RMSE (0.1253). This indicated that regularisation alone, without removing collinear variables, offered negligible practical improvement.

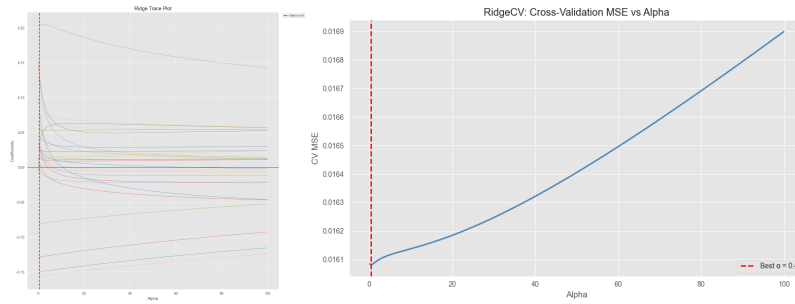


Figure 4: Ridge Trace and RidgeCV Plot

2.3 Model M₃: Forward-Selected OLS on Reduced Predictors

Given the multicollinearity findings, the following variable removal strategy was adopted: (i) *lease_commence_date* was dropped in favour of *remaining_lease_years*, as the two are near-perfectly collinear and *remaining_lease_years* is the more directly interpretable lease metric; (ii) *flat_type* was dropped because it is structurally associated with *flat_model* (chi-squared $p < 0.05$), and *flat_model* carries richer model-specific information.

Forward stepwise selection (minimising AIC at each step) was then run over the remaining candidate predictors: *floor_area_sqm*, *flat_model*, *region*, *remaining_lease_years*, and *average_storey*. The selection procedure added features in the following order (Table 5):

Step	Added Feature	AIC	Adj. R ²
1	<i>floor_area_sqm</i>	-65.3098	0.4923
2	<i>flat_model</i>	-875.7308	0.6371
3	<i>region</i>	-1583.4216	0.7285
4	<i>remaining_lease_years</i>	-2758.4484	0.8319
5	<i>average_storey</i>	-3098.1051	0.8537

Selected features: ['floor_area_sqm', 'flat_model', 'region', 'remaining_lease_years', 'average_storey']

Table 5: Forward selection steps with AIC and adjusted R².

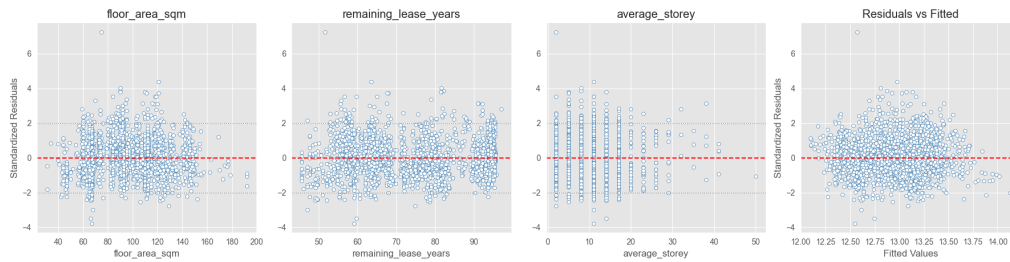


Figure 5: Residual plots for model 3

Residual diagnostics for M₃ confirm that the model assumptions are reasonably satisfied. The residual-vs-predictor plots for the three numerical features (*floor_area_sqm*, *remaining_lease_years*, *average_storey*) show no systematic pattern, supporting linearity. The mild heteroscedasticity pattern observed in M₀ persists but is slightly attenuated. No Cook's Distance exceeded 1, confirming the absence of unduly influential observations. VIF values are substantially reduced in M₃ (Table 4), with the highest being 65.3 for *remaining_lease_years*, a meaningful improvement over M₀, though moderate collinearity remains an acknowledged limitation.

2.4 Model M₄: Weighted Least Squares (IRLS)

The mild heteroscedasticity identified in M₀ and persisting in M₃ motivated the exploration of Weighted Least Squares (WLS) as a remediation strategy. WLS was applied to the same predictor set as M₃

(*flat_model*, *region*, *floor_area_sqm*, *remaining_lease_years*, *average_storey*). Weights were estimated via Iteratively Reweighted Least Squares (IRLS): at each iteration, the squared OLS residuals were regressed on the predictors to model the variance function, and observation weights were set to the reciprocal of the fitted variance. Weights were capped at the 99th percentile to prevent explosive upweighting of near-zero variance observations.

The IRLS procedure did not converge within 100 iterations, as evidenced by the persistently large maximum weight changes (approaching 10^6) across all iterations. This indicates that the variance function could not be stably estimated from the data, likely because the heteroscedasticity in the log-transformed response is too mild and irregular to support a reliable parametric variance model via OLS on squared residuals. Consequently, M_4 did not yield a stable or meaningfully improved fit over M_3 , and the WLS approach was not pursued further. The final selected model remains M_3 , which balances goodness-of-fit, parsimony, and interpretability without requiring non-convergent iterative weighting.

2.5 Final Model Interpretation

Quantitative predictor: floor_area_sqm: Holding all other variables constant, each additional square metre of floor area is associated with an increase of approximately **0.0105** in log resale price. This aligns with the intuitive expectation that larger flats command higher prices, independent of flat type or model.

Categorical predictor: region: Relative to the Central region, flats in the North-East and East regions are associated with approximately **0.296** and **0.224** lower log resale prices respectively. This reflects the well-documented price premium of centrally located HDB flats, driven by proximity to the CBD, amenities, and transport infrastructure.

Conclusion

This report proposed and evaluated several linear models for predicting HDB resale prices in Singapore using July 2020 transaction data. Beginning with a full OLS model M_0 (adjusted $R^2 = 0.859$), we identified severe multicollinearity, particularly between *lease_commence_date* and *remaining_lease_years*, and between *flat_type* and *flat_model*. Ridge regression provided negligible improvement. The final model M_3 , selected via forward AIC minimisation on a reduced predictor set, regresses $\ln(\text{resale_price})$ on *flat_model*, *region*, *floor_area_sqm*, *remaining_lease_years*, and *average_storey*. M_3 attains an adjusted R^2 of 0.854 with substantially reduced multicollinearity and superior parsimony, and is identified as the best linear model for this dataset.

M_3 achieves a comparable adjusted R^2 to M_0 (0.854 vs 0.859) with 7 fewer parameters, substantially reduced multicollinearity, and better parsimony. The slight AIC increase is an acceptable trade-off for a model with far less collinearity and superior interpretability.

The fitted equation of the final model M_3 is therefore:

$$\begin{aligned} \ln(\hat{\text{resale_price}}) = & 11.1998 + 0.0942\mathbf{1}_{\text{Apartment}} + 0.5028\mathbf{1}_{\text{DBSS}} + 0.1329\mathbf{1}_{\text{Improved}} + 0.2022\mathbf{1}_{\text{Maisonette}} + 0.1372\mathbf{1}_{\text{ModelA}} + 0.3489\mathbf{1}_{\text{ModelA-Maisonette}} + 0.1419\mathbf{1}_{\text{ModelA2}} \\ & + 0.2553\mathbf{1}_{\text{NewGeneration}} + 0.1548\mathbf{1}_{\text{PremiumApartment}} + 0.3525\mathbf{1}_{\text{PremiumApartmentLoft}} + 0.2462\mathbf{1}_{\text{Simplified}} + 0.0546\mathbf{1}_{\text{Standard}} + 0.8369\mathbf{1}_{\text{Terrace}} + 0.4736\mathbf{1}_{\text{TypeS1}} - 0.2243\mathbf{1}_{\text{East}} - 0.4339\mathbf{1}_{\text{North}} \\ & - 0.2956\mathbf{1}_{\text{North-East}} - 0.3549\mathbf{1}_{\text{West}} + 0.0105 \cdot \text{floor_area_sqm} + 0.0098 \cdot \text{remaining_lease_years} + 0.0095 \cdot \text{average_storey} \end{aligned}$$